# Fairness-aware Analysis of Centrality Measures in Social Networks

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**Abstract.** In real-world social networks, homophily drives community formation, often reinforcing structural inequalities. These networks typically follow a power-law distribution in community sizes, leading to dominant majorities and marginalized minorities. Centrality measures, widely applied in influence propagation, marketing, and resilience studies, are key tools for identifying influential nodes, and it is important to study disparities in these measures. This work systematically examines the fairness of local, semi-local, global, and modular centrality measures across both synthetic and real-world homophilic networks. Using uniform and non-uniform Independent Cascade Models (ICM) to simulate information diffusion, we analyze how influential nodes based on different centrality measures spread influence across different groups. Our results reveal that the majority groups consistently gain disproportionate influence, especially under non-uniform ICM. We further find that fairness varies a lot across individual centralities, regardless of their broader category, with fairness-aware pageranks consistently achieving the most equitable results.

**Keywords:** Fairness · Centrality Measures · Influence Propagation

## 1 Introduction

Social networks represent complex systems where individuals (nodes) interact through various relationships (edges), such as friendship, collaboration, or online communication. Analyzing these networks provides valuable insight into the formation of relationships, the structural organization of communities, and the influence of individual nodes on the overall system. However, real-world social networks often exhibit structural inequalities driven by social, economic, and political factors that shape their evolution. Two key mechanisms influencing network formation are (i) homophily - the tendency of individuals to connect with similar others, leading to the formation of communities [19,28], and (ii) preferential attachment, where well-connected high-degree nodes are more likely to gain new connections [2]. Social networks often originate with several homophilic

factors; for example, gender-based homophily often results in male users forming denser and more influential clusters [31,18]. The homophily coupled with preferential attachment can lead to the glass ceiling effect, limiting opportunities for minorities, despite having comparable characteristics [1]. In real-world networks, community sizes follow a power-law distribution, where a few large communities dominate and coexist with many smaller communities [20]. If such structural inequalities are not addressed while designing network analysis methods, algorithmic decisions may become biased, specifically against minorities [25]. In response, fairness-aware algorithms have been proposed in the past few years for various network analysis tasks, including link prediction [24], influence maximization [32], misinformation mitigation [23], and centrality measures [33].

In this work, we focus on centrality measures, fundamental tools in social network analysis used to quantify the importance of nodes based on their structural position [30]. These measures are widely used in applications such as influence maximization, marketing, and recommendation systems. Centrality measures can be categorized as: (i) *local* (based on a node's immediate neighborhood), (ii) *semi-local* (incorporating limited extended information beyond direct neighbors), and (iii) *global* (utilizing entire network structure) [27]. Another class, *modular centrality*, incorporates community-level information when ranking nodes. However, most traditional centrality measures are fairness-agnostic, often reinforcing the influence of already dominant nodes and exacerbating existing disparities. To address this, Tsioutsiouliklis et al. [33] proposed two fairness-aware PageRank variants: (i) *Fairness-Sensitive PageRank*, which adjusts the jump factor to promote underrepresented groups, and (ii) *Locally Fair PageRank*, which imposes a fairness constraint locally at each step.

We investigate the fairness-performance trade-off in centrality measures across both synthetic and real-world homophilic networks. We simulate information diffusion using uniform and non-uniform Independent Cascade Models (ICM) to evaluate how centrality affects the spread across majority and minority communities. Our findings show that majority groups dominate information flow in homophilic networks, with disparities intensifying under non-uniform ICM. We observe that fairness varies with specific centrality choices than across their general categories, with fairness-aware PageRank variants exhibiting more equitable outcomes. This study contributes to the growing field of fairness-aware network analysis, shedding light on the biases in traditional centrality measures and offering insights into designing fairer influence-ranking measures.

## 2 Methodology

This section outlines the centrality measures, influence propagation models, datasets, and evaluation metrics used in our experimental analysis.

### 2.1 Centrality Measures

This work examines 25 centrality measures grouped into three categories: local, semi-local, and global, with modular centralities highlighted in bold in Table 1.

| Type | Name |
|---|---|
| Local | Degree [9], H-index [13], $H^2$-index [26]<br>Z-score[14], Clustering Coefficient (CC) [4] |
| Semi-local | Chen *et al.*(CLSZZ) [7], Local Structural Centrality(LSC) [10]<br>Local RASP with Extended Neighbourhood Concept(LASPN) [34] |
| Global | K-shell with gravity(KSG) [35], Bridging [22]<br>Flow Betweenness(FB) [5], K-core [29], Katz [16]<br>Flow Closeness (FC) [5], **Participation Coefficient (PC)** [22]<br>Harmonic [21], **Fairness-Sensitive PageRank (FSPR)** [33]<br>PageRank [6], **Locally Fair PageRank (LFPR)** [33]<br>Betweenness [8], **Clustering Based Mediator (CBM)** [22]<br>Closeness [9], **Modulus Modular Betweenness (MMB)** [11]<br>Eigenvector [3], **Modulus Modular Closeness (MMC)** [11] |

Table 1: Centrality Measures (modular centrality measures are in bold)

Local centrality measures compute a node's importance based on its local neighborhood, while global centrality measures use the entire network. Semi-local centrality measures consider a node's neighbors and their neighborhoods, capturing a limited structural reach. Modular centrality incorporates community-level information, assessing a node's role both within and across communities.

These measures are used to rank nodes in real-world networks, and their effectiveness is evaluated through various applications, such as influence maximization, influence minimization, and network robustness. In this study, we focus on both effectiveness and fairness within the influence maximization framework [17]. Given a budget $k$, the top-$k$ nodes identified by a centrality measure are selected as seed nodes to initiate information diffusion across the network.

### 2.2 Influence Propagation Model

For simulating information propagation, we use the Independent Cascade Model (ICM). In ICM, each node is either *influenced* or *susceptible*. Initially, all nodes are susceptible, and a set of seed nodes is marked as influenced. In each iteration, each node $u$ influenced in the last iteration attempts to influence each of its susceptible neighbors $v$ with an influence probability $I_{uv}$. The newly influenced nodes continue propagating influence in subsequent iterations. The process terminates when no new nodes are influenced.

In real-world influence propagation, influence probabilities often differ between intra- and inter-community edges [12]. In our experiments, we fix the intra-community probability at 0.1 and vary the inter-community probability as (0.1, 0.05, 0.025), defining three models: $ICM_1$, $ICM_2$, and $ICM_3$, corresponding to uniform ($ICM_1$) and non-uniform ($ICM_2$, $ICM_3$) propagation.

### 2.3 Dataset

To analyze the fairness-performance trade-off of different centrality measures, we conduct extensive experiments on both synthetic and real-world networks.

Synthetic networks are generated using the Homophily BA model [15], which extends preferential attachment by introducing hyperparameters to control homophily and community size ratio. For our experiments, we generated synthetic networks with 10,000 nodes divided into two communities (8,000 nodes in majority and 2,000 in minority), and an average degree of 4. For each homophily level in $\{0.2, 0.4, 0.6, 0.8\}$, we created 10 independent networks which are used to compute average and standard deviation shown in the results. Additionally, we considered two real-world networks: Facebook and DBLP. The Facebook network consists of $3,232$ majority and 807 minority nodes; having a total $4,039$ nodes and $88,234$ edges. The DBLP network comprises two communities with $12,259$ and $4,242$ nodes; containing $16,501$ nodes and $66,613$ edges.

### 2.4 Evaluation Metrices

Given a network $G = (V, E)$ with $C = \{c_1, c_2, \cdots, c_m\}$ representing $m$ communities, and a seed set $S$ containing the initially chosen influencers, we evaluate the effectiveness of different centrality measures based on their influence outreach and fairness. The *Influence Outreach* $I(S, G)$ shows the total number of influenced nodes at the termination of the influence propagation process started by seed nodes $S$.

**Fairness Metrics.** Let $I(S, c_i)$ be the influence outreach of seed nodes $S$ within the community $c_i$. Fairness is assessed using the following two metrics:

- Disparity: It is the maximum normalized outreach gap across all communities [25], and is computed as:

$$Disparity(G, S) = \max_{i,j \in \{1,2,\ldots,|C|\}} \left| \frac{|I(S, c_i)|}{|c_i|} - \frac{|I(S, c_j)|}{|c_j|} \right|$$

  The score ranges from 0 to 1, where 0 represents fair outreach, and as the score moves towards 1, it indicates an increasingly unfair case.
- Influence Balance Ratio (IBR): The IBR quantifies skewness by comparing the influence outreach between the largest and smallest communities. It is calculated as:

$$IBR(G, S) = \frac{I(S, argmax_{c_i \in C} |c_i|)}{I(S, argmin_{c_i \in C} |c_i|)}$$

  A score of 1 indicates that the centrality is fair, whereas a score higher than 1 suggests that the centrality favors the majority and otherwise the minority.

## 3 Results and Discussion

To evaluate the fairness of different types of centrality measures, we first conduct experiments on HBA networks having different homophily using both uniform ($ICM_1$) and non-uniform Independent Cascade Models ($ICM_2$, and $ICM_3$) (as

described in Section 2.2). All centrality measures are used to identify a seed node set of size 100 for spreading influence, and their performance is evaluated using the fairness metrics and influence outreach.

Figures 1 and 2 present Disparity vs. Influence Outreach and IBR vs. Influence Outreach, respectively, across varying levels of homophily (0.2 to 0.8) and ICM models. Under the uniform model ($ICM_1$), higher homophily leads to reduced outreach and increased disparity and IBR. In contrast, non-uniform models ($ICM_2$, $ICM_3$) show increases in both influence outreach and bias, as shown by disparity and IBR. However, fairness-aware PageRank exhibits a dis-

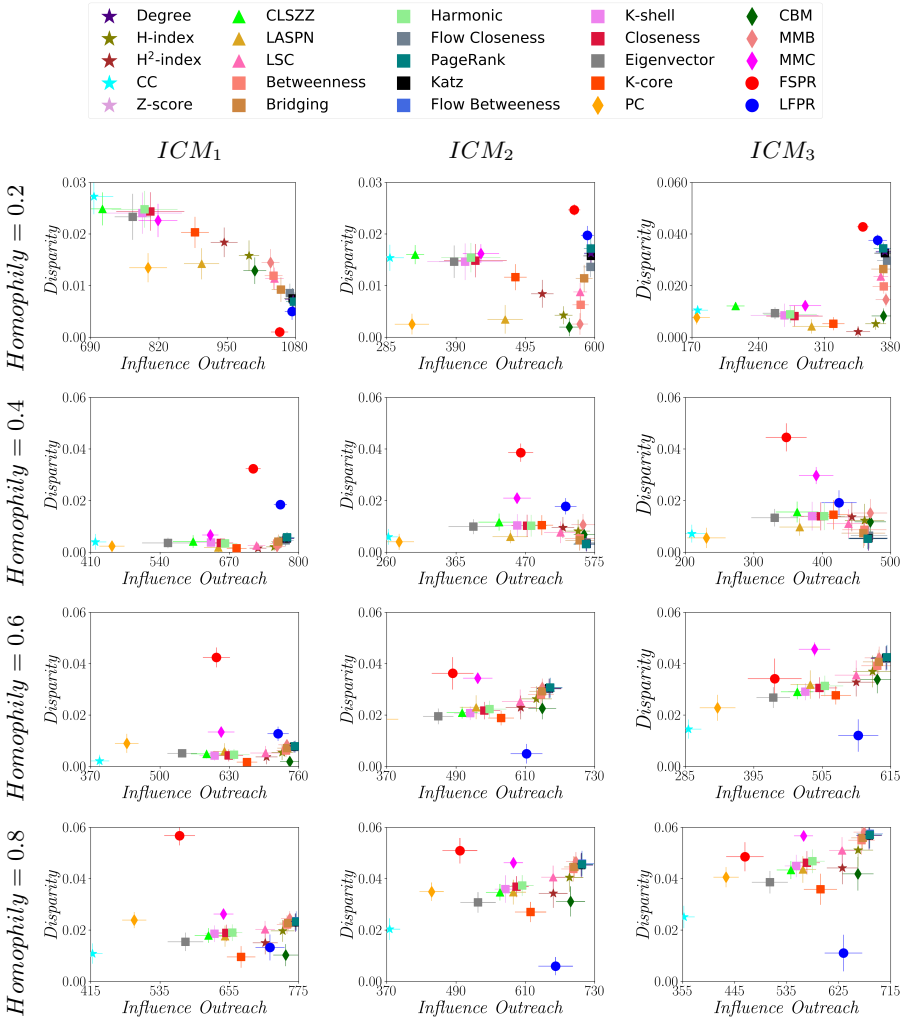



Fig. 1: Disparity versus Influence Outreach for uniform and non-uniform ICMs on HBA networks across varying homophily levels.

tinct pattern, where disparity continuously decreases as the network becomes more homophilic, because these measures explicitly redistribute influence toward underrepresented nodes, mitigating the dominance of the majority that intensifies with increasing homophily. Moreover, the LFPR (locally fair PageRank) has higher outreach and lower disparity than the FSPR (Fairness-Sensitive PageRank). We also observe that different classes of centrality do not have a specific bias pattern (as shown by disparity), and both influence outreach and fairness vary significantly among individual measures within each category: local, semi-local, global, and modular. Different semi-local centralities show varying

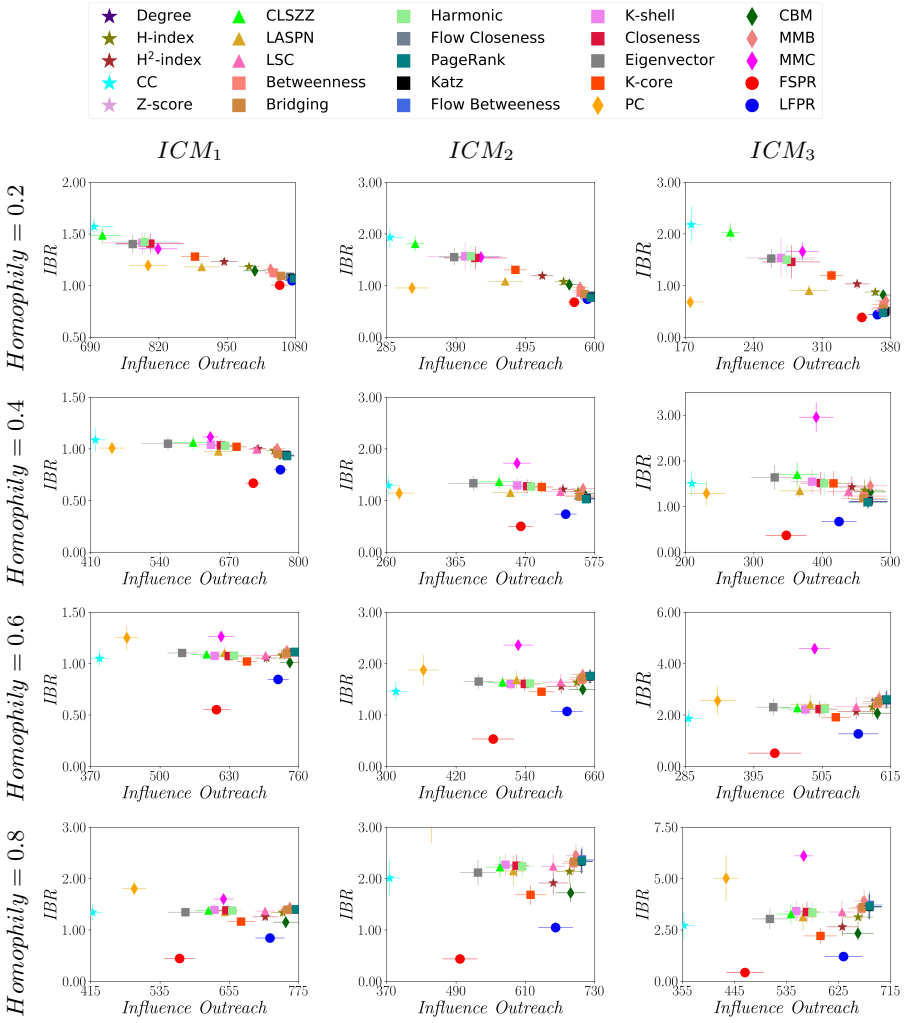



Fig. 2: IBR versus Influence Outreach for uniform and non-uniform ICMs on HBA networks across varying homophily levels.

performances in terms of influence outreach and fairness. For example, in low-homophilic networks, LSC achieves higher outreach and lower disparity. However, as homophily increases, its bias grows, as more seeds are selected from the majority group, and influence spreads predominantly within it due to increased density. Centrality measures providing higher outreach include Katz, PageRank, Flow-Betweenness, Degree, and Flow-Closeness.

IBR plots provide more insights with respect to how influence is distributed among minorities and majorities. We observe that for homophily 0.2, there are a few centralities, including Degree, Betweenness, Z-score, LSC, Bridging, PageRank, Katz, Flow-Betweenness, FSPR, and LFPR, that tend to influence the minority group more than the majority. This behavior becomes more prominent under non-uniform ICMs, especially in the extreme non-uniform influence propagation setting ($ICM_3$ in our case). The reason is that the centralities, like Betweenness, Bridging, Flow-Betweenness, choose nodes lying on the border of communities, and therefore, in the end, the fraction of nodes influenced in the minority community is higher than the majority, as the minority community is smaller in size and has more influential connections with such selected seed nodes than the majority. As we keep increasing homophily, the spread of influence in the majority increases much faster than in the minority. In a high homophilic network (homophily 0.8), for the uniform ICM model, only FSPR and LFPR impact minorities more, and this effect for LFPR disappears for the non-uniform models, as in those cases, the majority is more influenced than the minorities if seed nodes are selected using locally fair PageRank. Overall, in homophilic networks, with a realistic influence propagation model, i.e., non-uniform cascade models, the majority always benefits more than the minority.

As shown in Fig.1 and Fig.2, all centralities increasingly favor majority nodes in more homophilic networks under the non-uniform ICM model. Path-based centralities, such as Betweenness and Bridging, become less effective in identifying key nodes fairly as homophily increases, since dense intra-community connections within the majority limit cross-community paths and reduce their ability to capture fair influential bridging nodes. Similarly, the Closeness centrality favors majority nodes, as minority nodes are farther from the network core, limiting their chances of being selected as seeds. Eigenvector centrality remains stable but biased toward dense hubs, while PageRank favors nodes in majority communities due to its random walk nature. Harmonic centrality penalizes distant nodes, disadvantaging minority and bridging nodes. The K-shell, H-index, and $H^2$-index focus on neighborhood connectivity and tend to favor nodes in densely connected regions, making them less suitable for fair influence spreading in homophilic networks dominated by majority groups.

Overall, the Local Fair PageRank (LFPR) and Fair-Sensitive PageRank (FSPR) centralities outperform other measures in fairness metrics, as shown in Fig. 1 and Fig. 2. We further observe a trade-off that the LFPR achieves higher outreach, while the FSPR ensures higher fairness. These results highlight the importance of considering fairness-aware centrality rankings for such networks. Given the sparse literature on fair centrality, it will be beneficial to develop fair centrality

measures and test them across different types of networks with varying influence propagation models.

### 3.1 Impact of Seed Set Size

We further analyze how the influence outreach and fairness is impacted as we increase the number of seed nodes; results are shown in Fig. 3. We observe that as the number of seed nodes increases, disparity increases for most of the centrality measures as some centralities favour the majority and some favour the minorities (as discussed above). In highly homophilic networks, when the seed set is small, all centrality measures predominantly favor majority nodes. However, as the seed set size increases (to 100 in our case), FSPR gradually shifts toward favoring minority nodes. In contrast, within a less homophilic network, when the number of seed nodes is small, only LASPN favours the minority, while all other centrality measures continue to favour the majority. As the seed set expands to 100, the pattern changes significantly: FSPR, LFPR, Pagerank, Degree, Z-score, Bridging, LSC, Participation Coefficient, Betweenness, Flow-Betweenness, and MMB begin to favour the minority nodes, whereas the remaining centrality measures still prioritise the majority. In less homophilic networks, node disparity remains relatively stable. However, in more homophilic networks, higher disparity is observed for larger seed sets compared to less homophilic networks. In moderately homophilic networks (homophily = 0.4), disparity initially increases with the number of seed nodes but later decreases, as a larger seed set allows influence to spread more evenly across both communities.

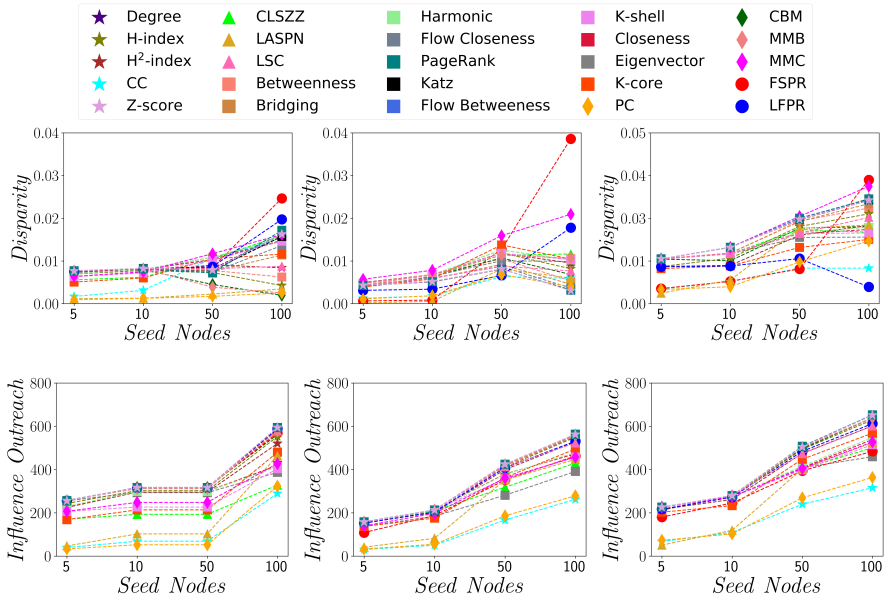
### 3.2 Fairness in Seed Nodes

We further analyze fairness at the level of selecting seed nodes using different centrality measures. We compute *Seed Difference*, i.e., the number of seed nodes selected from the majority community minus those from the minority. Fig. 4 illustrates the relationship between seed difference and influence disparity for various centrality measures on three homophilic networks with 100 seed nodes. The plot for the network with $h = 0.8$ is excluded due to space limitations, as it exhibits a trend similar to $h = 0.6$.

At low homophily (0.2), seed differences closer to zero yield lower disparities. Notably, three modular centrality measures: CBM, MMB, and PC perform well, with MMB achieving near-zero disparity with a slightly minority-biased seed selection. At moderate homophily (0.4), the disparity remains similar; however, all centrality measures favor the majority in seed selection, except for LFPR and FSPR. At higher homophily levels (0.6 and 0.8), disparity increases with increasing seed difference, as all centralities, except FSPR, select more seed nodes from the majority community. In contrast, FSPR shows higher disparity while favoring minority nodes in both seed selection and influence outreach. In highly homophilic networks (h=0.6), LFPR achieves the lowest disparity ($< 0.01$) and minimal bias in seed selection, clearly outperforming all other methods in terms of fairness.

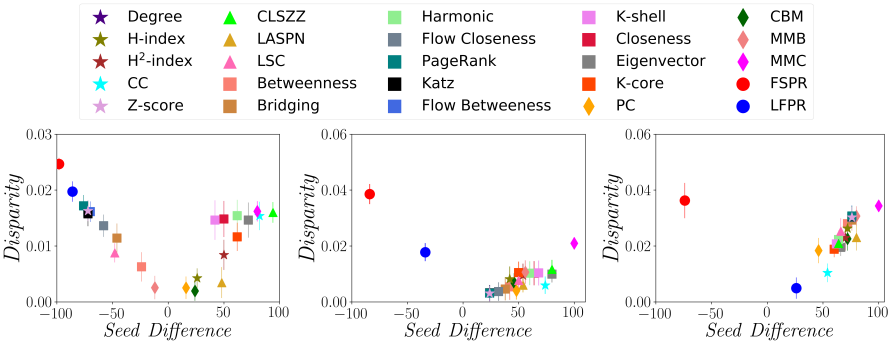
### 3.3 Results on Real-world Networks

Fig. 5 shows disparity versus influence outreach for various centrality measures on the Facebook and DBLP networks under the $ICM_2$ diffusion model for varying numbers of seed nodes (5, 10, 50). The average degrees of the FB and DBLP




Homophily = 0.2 Homophily = 0.4 Homophily = 0.6

Fig. 3: Disparity and influence outreach versus the number of seed nodes across different homophily values for $ICM_2$ model.




Homophily = 0.2 Homophily = 0.4 Homophily = 0.6

Fig. 4: Influence disparity and seed difference between majority and minority for various centrality measures under the $ICM_2$ diffusion model on HBA networks.

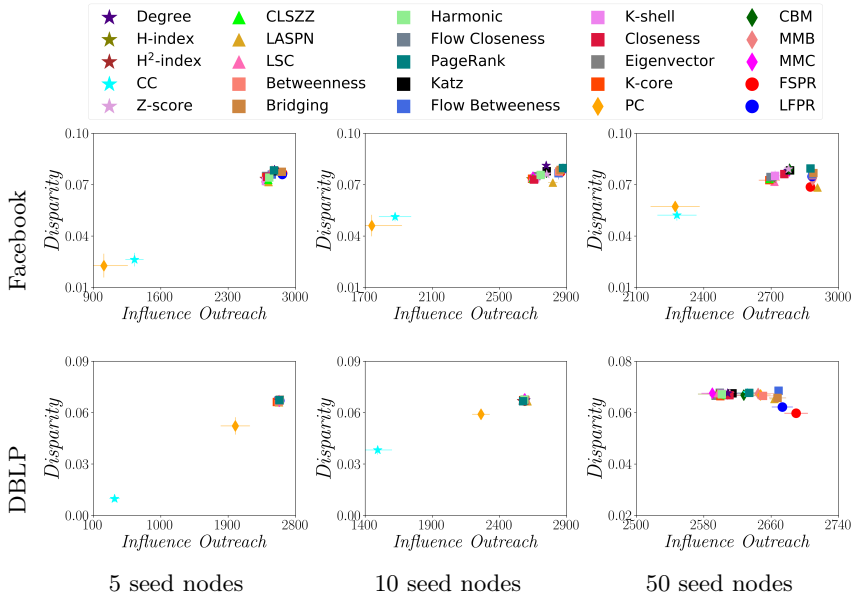



Fig. 5: Disparity versus Influence outreach for different centrality measures using the $ICM_2$ diffusion model on the Facebook and DBLP networks.

networks are 43.69 and 8.07, respectively, substantially higher than those of the synthetic networks. The results for 100 seed nodes closely match those for 50 as both has similar outreach due to high density, and are omitted due to space constraint. As Facebook is denser than DBLP, we observe that even with an increase in seed nodes, disparity and influence outreach remain stable for most centralities. In DBLP, the disparity tends to decrease as the size of the seed set increases. This suggests that with larger seed sets, influence spreads more evenly across groups, diminishing the dominance of structurally advantaged nodes. Our observations on real-world networks are consistent with synthetic networks. Interestingly, LFPR and FSPR exhibit a better disparity versus influence outreach trade-off in larger seed sizes, particularly on the DBLP network, indicating their potential for fairer influence spread on sparse homophilic networks.

These observations highlight the importance of selecting centrality measures that consider not only outreach efficiency but also fairness, particularly in contexts where equitable group representation is crucial.

## 4 Conclusion

This study systematically evaluated the fairness of 25 centrality measures, including local, semi-local, global, and modular centralities, across synthetic and real-world homophilic networks. Using both uniform and non-uniform Independent Cascade Models (ICM), we analyzed how centrality-based seed selection

affects influence outreach and group-level fairness. Results reveal that most traditional centralities favor majority communities, with this bias intensifying as homophily increases. While no specific category of centrality measures consistently ensures fairness, individual measures vary substantially in their behavior. Among all, the Local Fair PageRank (LFPR) and Fair-Sensitive PageRank (FSPR) demonstrate the best trade-off between fairness and influence outreach, with LFPR achieving higher outreach and FSPR offering greater equity where the seed nodes and the fraction of influenced nodes from the minority are larger. These findings underscore the need for fairness-aware adaptations of classical network measures. Future work should explore other influence propagation models, such as linear threshold and SIR models, to verify the findings in those cases.

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